

Haoxiang Wan

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Education

University of Houston, Houston, TX

Aug. 2025 – Expected May 2029

Ph.D. in Electrical Engineering

Advisor: [Prof. Xingpeng Li](#), Renewable Power Grid (RPG) Laboratory

University of Illinois at Urbana-Champaign, Urbana, IL

May 2025

Master of Engineering in Energy Systems

The University of Sheffield, Sheffield, UK

June 2024

Bachelor of Engineering in Electrical and Electronic Engineering, *First Class Honours*

Technical Summary

Ph.D. researcher specializing in power system operations, planning, and large-load integration. Experienced in SCUC, stochastic generation and transmission expansion planning, AC and DC hosting-capacity assessment, and GPU-accelerated full-year optimization on Texas-scale networks. Proficient in Python, Julia, Pyomo, JuMP, Gurobi, CPLEX, Ipopt, NVIDIA cuOpt/PDLP, and HPC computing.

Research Experience

Graduate Research Assistant

Aug. 2026 – Present

Renewable Power Grid (RPG) Laboratory, University of Houston, Houston, TX

Advisor: [Prof. Xingpeng Li](#)

- Developed a stochastic generation and transmission expansion model for ERCOT 2035 under uncertain data center energization, cutting expected cost by 26% against ERCOT's forecast-based plan.
- Showed that DC power flow screens overstate AC-verified data center hosting capacity at 113 of 123 buses of a synthetic Texas system.
- Built SCUC models that shift data center load across buses and hours to relieve congestion and reduce renewable curtailment.
- Quantified data center impact on co-optimized generation and transmission expansion with embedded unit commitment on IEEE RTS-24.

Research Aide (Summer Internship)

May 2026 – Aug. 2026

Argonne National Laboratory, Lemont, IL

Supervisor: [Dr. Jonghwan Kwon](#), Energy Systems and Infrastructure Analysis Division

- Built a full-year cross-stress framework for generation expansion planning under uncertain data center deployment, operating 30 candidate portfolios under 30 project-level deployment futures on a 254-county Texas 2030 system.
- Solved 1,161 full-year (8,760-hour) linear programs on GPUs with NVIDIA cuOpt/PDLP in Argonne's [A-LEAF](#) framework.

Publications

Published and Accepted

- **H. Wan**, L. Fang, J. Kwon, J. Silva-Rodriguez, J. Yang, J. Yellajosula, F. Jiang, and X. Li, "Impact of Security Modeling Fidelity on Hyperscale Data Center Hosting Capacity Assessment," to appear in *Proc. IEEE PES Grid Edge Technologies Conference & Exposition*, Salt Lake City, UT, USA, Apr. 2027.
- L. Fang, **H. Wan**, A. Srivastava, and X. Li, "Service-Aware Hosting Capacity Assessment of Edge Data Centers with Workload Flexibility," to appear in *Proc. IEEE PES Grid Edge Technologies Conference & Exposition*, Salt Lake City, UT, USA, Apr. 2027.
- **H. Wan**, J. Kwon, and X. Li, "Impact of Data Center Load on Co-Optimized Generation and Transmission Expansion Planning with Embedded Unit Commitment," to appear in *Proc. IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, Feb. 2027.
- **H. Wan** and X. Li, "Data Center Spatio-Temporal Load Flexibility in Security-Constrained Unit Commitment for Enhanced Grid Efficiency and Reliability," to appear in *Proc. IEEE Industry Applications Society (IAS) Annual*

Meeting, Vancouver, BC, Canada, Oct. 2026. [arXiv]

- **H. Wan**, L. Fang, and X. Li, “Grid Operational Benefit Analysis of Data Center Spatial Flexibility: Congestion Relief, Renewable Energy Curtailment Reduction, and Cost Saving,” in *Proc. IEEE Power & Energy Society General Meeting (PESGM)*, Montreal, QC, Canada, Jul. 2026. [arXiv]

In Preparation

- **H. Wan**, J. Kwon, S. Cho, S. B. Sharma, T. Levin, and X. Li, “Risk-Aware Selection of Generation Expansion Portfolios under Uncertain Data Center Deployment: A Full-Year Cross-Stress Framework,” in preparation for *Applied Energy*, 2026.
- **H. Wan** and X. Li, “Hedging an Uncertain Data Center Queue: Stochastic Generation and Transmission Expansion Planning in ERCOT,” in preparation for *IEEE Transactions on Power Systems*, 2026.
- **H. Wan**, J. Silva-Rodriguez, J. Kwon, and X. Li, “AC Verification of DC Power Flow Screens for Data Center Hosting Capacity and Siting,” in preparation for *IEEE Power & Energy Society General Meeting (PESGM)*, 2027.
- **H. Wan**, I. Castillo, and X. Li, “Impact of Wind-Solar-Battery Growth on Electricity Prices: Reserve Scarcity and Transmission Congestion,” in preparation for *IEEE Gulf Coast Conference on Energy, Communications and Automation (GECA)*, 2027.

Teaching Experience

Graduate Teaching Assistant

Jan. 2026 – May 2026

ECE 5380 Power Electronics & Electric Drives, University of Houston

Instructor: [Dr. Hao Huang](#)

- Graded homework and examinations and provided written feedback for a synchronous online graduate course on power semiconductor devices, converters and inverters, and motor drives.

Graduate Teaching Assistant

Aug. 2025 – Dec. 2025

ECE 5388 Renewable Energy Technology, University of Houston

Instructor: [Dr. Hao Huang](#)

- Assisted with instruction and assessment for a course on solar, wind, and grid integration of renewable resources.

Technical Skills

- **Optimization & Modeling:** Pyomo, JuMP, AMPL, Gurobi, CPLEX, Ipopt, NVIDIA cuOpt/PDLP, MadNLP
- **Power System Tools:** PowerModels.jl, PyPSA, MATLAB/Simulink, LTspice
- **Programming Languages:** Python, Julia, MATLAB, C, C++
- **Machine Learning:** PyTorch, scikit-learn, pandas, NumPy
- **Computing:** Linux/HPC (Slurm, PBS), GPU computing (CUDA), Git
- **Languages:** English (fluent), Mandarin Chinese (native)

Honors and Awards

- Engineering Excellence Scholarship, The University of Sheffield 2021–2022, 2022–2023
- NCUK Sheffield Scholarship, The University of Sheffield 2021–2022, 2022–2023

Professional Affiliations

Student Member, IEEE Power & Energy Society (PES) and IEEE Industry Applications Society (IAS)